CCC-C-419D
September 28, 1973
SUPERSEDING
Fed. Spec. CCC-C-419C
July 29, 1968

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON, UNBLEACHED, PLIED-YARNS

ARMY AND NUMBERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for grey (unbleached) cotton duck cloth.

1.2 Classification.

1.2.1 Types. The cotton duck cloth shall be of the following types as specified (see 6.2):

Type I - Numbered duck.
Type III - Army duck.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276 - Thread, Cotton.
PPP-P-1135 - Packaging and Packing of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-Synthetic Fiber Blends).

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 191 - Textile Test Methods.
Fed. Std. No. 751 - Stitches, Seams, and Stitching.

FSC 8305

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Military Standard:

MIL-STD-105 - Sampling procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

PUBLICATION:

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Yarn. The yarn shall be cotton which has been carded, drawn, spun and twisted into plied yarn.

3.2 Physical requirements. Cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in tables I and II when tested as specified in 4.4.

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TABLE I. Physical requirements for type I, numbered duck

Number of duck, commercial designation	Weight (ounces/ sq.yd.) minimum	Yarn ply minimum		Yarn per inch, minimum		Breaking strength (lbs.) min.		Air permeability ($ft^3/min.ft^2$) maximum
		Warp	Filling	Warp	Filling	Warp	Filling	
Hard texture:								
12/0	48.00	9	9	20	12	800	430	-
8/0	40.00	8	14	26.5	12.5	750	600	-
2/0	31.90	5	7	24	16	465	435	-
1/0	30.31	5	6	24	16	450	405	-
1	28.71	5	5	26	19	440	370	2
2	27.12	5	4	26	19	420	345	-
3	25.53	4	5	29	20	390	330	-
4	23.93	4	4	31	22	375	300	2
5	22.33	3	4	35	23	345	285	-
6	20.74	3	3	35	25	335	250	2
8	17.55	3	3	43	25	285	210	2
10	14.35	3	3	45	27	245	160	4
11	12.75	3	2	45	34	240	140	-
12	11.16	2	2	45	35	195	120	4
Medium texture:								
1	28.71	5	5	26	15	425	345	-
2	27.12	5	4	26	15	410	320	-
3	25.53	4	5	29	15	370	315	-
4	23.93	4	4	29	17	350	290	-
5	22.33	3	4	34	18	315	285	-
6	20.74	3	3	34	18	305	250	-

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TABLE II. Physical requirements for type III, Army Duck

Weight (ounces/sq.yd.) minimum	Yarn ply, minimum		Yarn per inch, minimum		Breaking strength (lbs) minimum		Air permeability (ft3/min/ft^2 maximum
	Warp	Filling	Warp	Filling	Warp	Filling	
8.25	2	2	60	58	125	120	-
9.85	2	2	52	40	160	110	4
12.29	3	2	44	34	210	130	4
14.77	3	3	44	28	235	175	-
15.90	3	3	45	34	245	200	-
18.48	4	4	42	28	315	200	-

3.2.1 Width. The width of the cloth inclusive of selvages shall be as specified (see 6.2). The following tolerances shall be permitted:

Width of duck (inches)	Tolerance (inches)
40 or less	+1/4 -1/4
Over 40 to 51 inclusive	+3/8 -1/4
Over 51 to 81 inclusive	+5/8 -3/8
Over 81	+3/4 -3/8

3.2.2 Weave. The weave shall be plain, one up, one down.

3.3 Nonfibrous material content. The cloth shall contain not more than 2.5 percent starch and protein content excluding chloroform-soluble and water-soluble material. The total sizing, finishing and other nonfibrous material content of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.4.

3.4 Length and put up. Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 40 yards and shall be put up in rolls in accordance with PPP-P-1135. The maximum length of the rolls when converted to pounds shall not exceed the maximum weight specified in PPP-P-1135. For duck No. 10 and lighter in weight, pieces shall be sewn together with a butted FSa-1 type seam of FED-STD-751 using ticket No. 10, 3 ply unbleached cotton thread conforming to type IIIA of V-T-276, and having a minimum of 6 stitches per inch. The stitching shall be placed 1/2 to 5/8 inch from the cut or raw edge of the cloth. For cloth heavier in weight than No. 10, the pieces shall be lapped and not sewn together. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches of the roll on the outer surface.

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3.5 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.6 Face marking. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.7 Workmanship. The cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3 Examination of end item. Examination of the end item shall be in accordance with 4.3.1 through 4.3.3.

4.3.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.3.1.1 with the following exceptions: Glossary numbers 1, 2, 3, 5, 14, 26, 27, and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks

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shall be scored if they result in a clearly noticeable separation of warp yarns. Each knot or slub shall be scored as a defect only when it exceeds three times the normal yarn diameter. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 40 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 60 points. If one roll exceeds 60 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality examination. The lot shall be unacceptable if one or more rolls in the second sample exceed 60 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	-	one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

The following defects when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Width not within specified tolerance.
Objectionable odor.
Excessive neppiness.

4.3.2 Examination for length.

4.3.2.1 Individual rolls. During the yard-by-yard examination each roll in the sample shall be examined for put up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present.

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Defects

Any length found to be more than 2 yards less than the length marked on the roll ticket.
Any piece less than 40 yards in length.
For No. 10 duck and lighter in weight:
a. Pieces not sewn together with a butted FSa-1 type seam.
b. Stitches less than the minimum specified.
c. Stitching not 1/2 to 5/8 inch from the cut or raw edge of the cloth.
For duck heavier in weight than No. 10, pieces sewn instead of lapped. Number of pieces in roll not stenciled on last 6 inches of rolls on outer surface.

4.3.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.3.3 Examination for face markings and compliance with the Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have the face stamping missing from either or both ends, or face stamping on wrong side, or are not labeled, ticketed, or invoiced in accordance with the Textile Fiber Products Identification Act.

4.3.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1135 to determine whether packaging, packing and marking comply with section 5 requirements.

4.4 Testing of the end item. The methods of testing specified in Fed. Std. No. 191 wherever applicable as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1-1/2 yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirements specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Test methods

Characteristic	Requirement paragraph	Test method
Weight	3.2	5041
Yarn ply	3.2	Visual 1/
Yarns per inch	3.2	5050 2/
Breaking strength	3.2	5100
Air permeability	3.2	5450
Weave	3.2.2	Visual 1/
Starch and protein content excluding chloroform-soluble and water-soluble material	3.3	2611
Total sizing, finishing and other nonfibrous materials	3.3	2611

1/ One determination shall be made from each sample unit and the results reported as "pass" or "fail".

2/ Except that for type I, 8/0 Hard Texture, the results shall be reported to the nearest 0.1 of a yarn.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1135.

6. NOTES

6.1 Intended use. The cloths covered by this specification are primarily intended for use in fabrication of tentage and equipage items.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, number and weight of cloth required (see 1.2 and 3.2).
(c) Width required (see 3.2.1).
(d) Length, of other than specified (see 3.4).
(e) Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD, MI, MU
Navy - MC
Air Force - 82

User activities:

Army - AT, EL, ME
Navy - YD

Preparing activity:

Army - GL

Civil Coordinating Activities:

GSA-FSS
COM-NBS
HEW-HEW
JUS-FPI
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DOCUMENT IDENTIFIER AND TITLE
CCC-C-419D
CLOTH, DUCK, COTTON, UNBLEACHED, PLIED-YARNS ARMY AND NUMBERED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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